



JAVA SPRING BOOT

ROADMAP



— Step by Step to Become a Spring Boot Developer —



Learn → Build → Practice → Grow

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1. Core Java Refresher

- OOP Concepts
- Collections Framework
- Exception Handling
- Multithreading, Lambda Expressions

🕒 1–2 Weeks

You will learn

Strong Java foundation required for Spring Boot development.



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2. Spring Framework Basics

- IoC Container & DI
- Beans & Configuration
- Annotations (@Component, @Autowired, etc.)
- Spring Core, AOP Basics

🕒 1–2 Weeks

You will learn

Core concepts of Spring Framework.



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3. Spring Boot Fundamentals

- Introduction to Spring Boot
- Auto-Configuration
- Starters & Dependencies
- Spring Boot Project Structure

🕒 1 Week

You will learn

How Spring Boot simplifies and accelerates Spring development.



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4. Building RESTful APIs

- @RestController, @RequestMapping
- Path Variables, Request Parameters
- RequestBody, ResponseEntity
- Exception Handling in APIs

🕒 1–2 Weeks

You will learn

Build robust REST APIs with best practices.



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5. Data Access with Spring Boot

- Spring Data JPA
- CRUD Operations
- Repository Interfaces
- Custom Queries, Pagination & Sorting

🕒 1–2 Weeks

You will learn

Work with databases efficiently using JPA and Hibernate.



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6. Spring Boot Security

- Spring Security Basics
- Authentication & Authorization
- Password Encoding
- Securing REST APIs (JWT Basics)

🕒 1–2 Weeks

You will learn

Secure your applications with authentication & authorization.



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7. Validation & Global Exception Handling

- Bean Validation (Jakarta Validation)
- @Valid, Validation Annotations
- Global Exception Handling with @ControllerAdvice

🕒 3–5 Days

You will learn

Handle errors gracefully and validate inputs.



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8. Configuration & Profiles

- application.properties / application.yml
- Profiles (dev, test, prod)
- Environment Specific Configuration

🕒 3–5 Days

You will learn

Manage different environments effectively.



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9. Testing Spring Boot Applications

- Unit Testing with JUnit 5 & Mockito
- Integration Testing
- Test Slices (@WebMvcTest, @DataJpaTest)

🕒 1 Week

You will learn

Write unit & integration tests for reliable applications.



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10. Projects & Deployment

- Build Real-World Projects
- API Documentation with Swagger/OpenAPI
- Deploy to Cloud (AWS / Render / Railway)
- Docker Basics (optional but recommended)

🕒 2–3 Weeks

You will learn

Build, document and deploy production-ready Spring Boot applications.



TOOLS & TECHNOLOGIES



Java



Spring Boot



Maven



MySQL



Git



Postman



Swagger

PROJECT IDEAS

- ✓ Employee Management System
- ✓ E-commerce REST API
- ✓ Blog Management System
- ✓ Booking / Reservation System
- ✓ Task Management System

TIPS FOR SUCCESS

- ✓ Practice daily
- ✓ Build projects
- ✓ Read documentation
- ✓ Solve real-world problems
- ✓ Stay consistent & keep learning



Consistent learning + Hands-on practice = Mastery in Spring Boot

